

Hongyang Du

11 East Manning St, Providence, RI | 240-413-5747 | hongyangdu182@gmail.com | Website | LinkedIn | Github

Image/Video Diffusion Models

World Models

Representation Learning

EDUCATION

Brown University

M.S. Computer Science (AI/ML Track)

Aug 2025 – May 2027

GPA 4.0/4.0

University of Maryland, College Park

B.S. Computer Science (Honors), Minor in Mathematics

Aug 2021 – May 2025

GPA 4.0/4.0

PROFESSIONAL EXPERIENCE

Adobe-Machine Learning Engineer Intern | Firefly Content AI Team

May 2026 – Present

- Architected a self-evolving agentic image generation system, featuring a structured skill memory that continuously distills workflows into reusable skills using learning signals from synthetic LLM priors and real user trajectories.
- Engineered a closed-loop skill evolution mechanism (insert/revise/deprecate) driven by multi-turn sessions. Curated memory mutations via a two-level multi-agent framework to address both short/long-tail conversations.
- Deployed a replay-gated strategy for safe skill bank updates, requiring counterfactual improvement on offline trajectories before advancing candidates to stable production via shadow/canary evaluations on live traffic.

iFlytek-Machine Learning Engineer Intern | Agriculture MLLM Team

June 2024 – Aug 2024

- Pretrained a 2B Chinese LLM from scratch based on the LLaMA2 architecture (e.g., RoPE positional encoding, RMSNorm, SwiGLU). Employed SentencePiece Unigram tokenizer to enhance Chinese linguistic modeling.
- Combined ResNet hierarchical classification and YOLOv10 disease-spot detection to classify over 1,000 crop diseases and pests. Constructed RAG-enhanced prompts using visual clues and a knowledge graph.
- Finetuned the model using LoRA and reinforcement learning for domain-specific adaptation. Build a Chinese-centric pretraining pipeline with support for FSDP, gradient checkpointing, and customized preprocessing for Chinese text.

VISUAL REPRESENTATION & GENERATION

Unified Representation Autoencoder for Image Diffusion Under Review

Present

- Conventional Representation Autoencoders (RAEs) exhibit a structural discrepancy, as reconstruction overfits to pixel features while diffusion overfits to deep semantic features, leaving cross-frequency mutual information isolated.
- We propose a self-supervised training paradigm that replaces heuristic feature-fusion, mathematically proven to capture joint correlations between pixel-level details and semantic features.
- By harmonizing reconstruction and generation in an optimized latent space, a unified decoder improves reconstruction fidelity and accelerates DiT convergence by 2x with lower gFID.

Video World Model

Feb 2026

ICML 2026, CVPR VGBE Challenge [3rd Place] & Workshop [Oral]

- Video Diffusion Models inherently struggle with temporal inconsistency and spatial drift, largely due to a lack of robust grounding in 3D structural priors and physical geometric laws.
- We proposed a self-supervised data framework to distill 3D priors from a Geometry Foundation Model (VGGT). Leveraging Direct Preference Optimization (DPO) for physical geometric laws alignment.
- With minimal preference data and RL fine-tuning, our framework significantly outperforms state-of-the-art baselines in 3D consistency and motion coherence, achieving superior physical plausibility on complex real-world benchmarks.

Implicit Neural Representation for Video Compression

May 2025

WACV 2026

- Conducted empirical study of video compression methods based on INRs. Developed a unified benchmarking platform to evaluate models across metrics PSNR, MS-SSIM, bitrate, training time, and decoding speed on HD datasets.
- Proposed a novel architecture RNeRV, which introduced a Weight Token Masking mechanism and HyperNetwork decoupling to achieve flexible bitrate control and efficient model deployment. Demonstrated +1.27% average improvement in visual quality under equal training budgets and enhanced real-time inference throughput.
- Built the XINC (eXplainable INR Compression) tool to visualize neuron-level activation and contribution maps across layers and time steps, improving interpretability and facilitating model optimization for video streaming.

Self-Evolving Vision Language Models From Zero Data Under Review

June 2026

- Proposed a novel training paradigm that eliminates the need for human-annotated data ("zero data") by designing an innovative three-role collaborative pipeline: Proposer, Coder, and Solver
- Empowered the model to autonomously construct visually complex scenes via code generation; implemented Group Relative Policy Optimization (GRPO) with role-specific reward mechanisms for sequential, closed-loop training.
- Drive a comprehensive performance leap across base models (e.g., Qwen3-VL, MIMO-VL) with significant gains in multimodal understanding, mathematical reasoning, and coding capabilities.

Agent Skill Dependency-Aware Structural Retrieval ACM CAIS Workshop 2026

Apr 2026

- Structured massive agent skill libraries as an offline dependency graph and retrieved small, relevant skill bundles at inference (seeding + PageRank under a token budget) instead of loading the full library.
- Reduced context tokens while improving downstream reward for LLM-based agents operating over large skill sets.

Vision Language Models Understanding and Hallucination NeurIPS 2025, CVPR 2025 Workshop [Oral]

Aug 2025

- Designed VideoHallu dataset for hallucination detection in synthetic video understanding; proposed 2 novel metrics (temporal consistency, semantic mismatch) to quantify alignment between captions and dynamic video scenes.
- Implemented robust hallucination detection framework integrating frame-level caption matching, embedding similarity (CLIP/BLIP), and cross-frame alignment; enhanced via GRPO post-training with curriculum learning on video QA datasets incorporating counterintuitive commonsense and physics reasoning.
- Conducted a systematic survey of cutting-edge vision-language models (e.g., GPT-4V, Gemini, Qwen-VL), detailing their model architectures, instruction alignment mechanisms, and evaluation protocols.

PUBLICATIONS

MM-Zero: Self-Evolving Multi-Model Vision Language Models From Zero Data

Z. Li*, H. Du*, C. Huang*, et al. · [EMNLP 2026](#)

VideoGPA: Distilling Geometry Priors for 3D-Consistent Video Generation

H. Du*, J. Ye*, X. Cong*, et al. · [ICML 2026](#)

Graph of Skills: Dependency-Aware Structural Retrieval for Massive Agent Skills

D. Liu*, Z. Li*, H. Du, et al. · [EMNLP 2026](#)

A Cookbook of 3D Vision: Data, Learning Paradigms, and Application

H. Du*, Z. Li*, D. Liu*, et al. · [CVPR 2026 Workshop](#)

How to Design and Train Your Implicit Neural Representation for Video Compression

M. Gwilliam, R. Zhang, N. Padmanabhan, H. Du, et al. · [WACV 2026](#)

VideoHallu: Evaluating and Mitigating Multi-modal Hallucinations for Synthetic Videos

Z. Li*, X. Wu*, Y. Qin, H. Du, et al. · [NeurIPS 2025](#)

A Survey of State of the Art Large Vision Language Models: Alignment, Benchmark, Evaluations and Challenges

Z. Li*, X. Wu*, H. Du, et al. · [CVPR 2025 Workshop \(Oral\)](#)

Ipelets for the Convex Polygonal Geometry

N. Parepally, A. Chatterjee, A. Gezalyan, H. Du, et al. · [SoCG 2024](#)

TECHNICAL SKILLS & AWARD

Languages: Python, Shell, Java, C++, Bash, Go, Rust, SQL, JS/HTML/CSS, OCaml, Lua, Racket, React, Matlab
Dev Stack: Spring Boot, Flask, Django, FastAPI, Node.js, RESTful APIs, Redis, MongoDB, Docker, Flutter, Next.js
AI/ML Stack: PyTorch, TensorFlow, Transformers, scikit-learn, OpenCV, LangChain, LlamaIndex, Haystack, NumPy
Awards: Robert Ma Scholarship, Break Through Tech Scholarship, Dean's List (7 semesters), AIME Qualifier (Top 5%), AMC 12 Top 1% Honor Roll of Distinction, TN State Math Competition 1st Place